

Amide activation: an emerging tool for chemoselective synthesis Daniel Kaiser, Adriano Bauer, Miran Lemmerer and Nuno Maulide * Chem. Soc. Rev., 2018, 47, 7899

SI No,	Name	Aspects to study	Scheme/Section no
1	Rachel (251CHO03)	Nucleophilic addition to keteniminium ion, generation and reactions of keteniminium ion, Challenge in the formation of keteniminium ions. Keteniminium ions in formal [2+2 additions]: need to use triflic anhydride and collidine, advantage of the approach. Stereospecificity study, Application: Corey synthesis of prostaglandin intermediate	Schemes-3,4 Scheme 5, 6 Scheme-9
2	Syeda (251CHO02)	Addition of Nucleophiles to keteniminium ions: Intermolecular addition of carbon nucleophiles Movassaghi synthesis of substituted pyridines (addition of Nucleophiles to secondary amides). Wang's indole synthesis employing EDA	Scheme-13 mechanism Scheme 14
3	Harshitha (251CHO04)	Reduction of activated amides	Scheme 19
4	Akash (251CHO07)	N-pyridinylation: Intermolecular addition of hetero atom Nucleophiles (relate to scheme 17a)	Scheme 22
5	Lavanya (251CHO15)	Umpolung functionalisation of amides Functionalization of thioamides	Scheme-28 Scheme 29
6	Chaitra (251CHO16)	Privileged amides: Structures of Weinreb and Szostak amides, anilides -special features and advantages Full form /short form/ structure and application of TBDPS (p 7905), DTBMP (p 7905, 7910 and many more pages) Hantzsch ester (p7906, 7908), EDA (p 7906), collidine (p 7902), lutidine-N-oxide, EtO ₃ BF ₄ (Meerwein salt, p 7908), Tf ₂ O (p 7902).	Scheme 47 (a), (b) and (c)
7	Suresh (251CHO10)		

Nonenzymatic Reactions in Natural Product Formation Leah M. Bouthillette, Victor Aniebok, Dominic A. Colosimo, David Brumley, and John B. MacMillan* Chem. Rev. 2022, 122, 14815–14841

SL NO	Name	Aspects to study	Scheme
1	Anna Rose (251CHO06)	Influence of biosynthetic studies Foundation of natural product biosynthesis (meaning of primary and secondary metabolites with ex). Meaning of BGC and its importance	Section 1.1, 1.2,
2	Simra Sardar (251CHO11)	Nonenzymatic reactions in cellular milieu (mainly in metabolic steps). Incomplete categorization of nonenzymatic processes in natural products (mainly in metabolic steps)	Section 1.3 Section 1.4
3	Aileen (251CHO14)	Categories of nonenzymatic mechanism	Fig. 1
4	Smirti (251CHO01)	Intramolecular cycloaddition reactions	Sec 2.1, Fig 2, 3 and 4
5	Sadashiv (251CHO12)	Multicomponent reactions: (meaning of multicomponent reactions) Homologous multicomponent dimerization reactions; meaning and examples	Sec 2.2 Sec 2.2.1, Fig 5, 7, 9, 12
6	Sreelaxmi (251CHO13)	Tailoring reactions; meaning, classes Investigation of stealthins Light induced reactions: Production of flueggedine Experimental validation of nonenzymatic [2 + 2] cycloaddition of dactylone to form aplydactylone by biological UV light	Sec 2.3. fig 23, 24, Fig 26 (b and c only) Sec 2.4 P. 14832 and Fig 10 (b) Fig 28, 29 a and b Fig 30, Fig 31
7	Pauline (251CHO05)	Discussion of common functional groups and precursors Role of anthranilic acid in the formation of a variety of nonenzymatically derived natural product scaffolds Representation of different potential biological applications of nonenzymatic mechanisms that create new molecular scaffolds.	Sec 4 Fig 32 Fig 33

